

Cell Design Specification — V11 (t1_r3)

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Case: t1_r3 — next-generation pure electric sedan battery (ED ≥ 392.61 Wh/kg, 4C fast charge without plating, $T_{\max} \leq 60\text{ }^{\circ}\text{C}$, overcharge 4.7 V without thermal runaway)

Date: 2026-08-26

Caliber note: all values mechanically taken from the Chen2020 parameter set, the V11 parameter file, and simulation outputs (``runs/exp/t1_r3/cell/r5_v11_*.json``); per-line sources annotated. Cell-level virtual design; engineering manufacturing drawings (tolerances) and line process cards are outside the pure-simulation boundary.

1. Basic Specification

Item	Value	Source
Electrochemical system	NMC811 (LG M50 calibration) / graphite	Chen2020 parameter set (OCP functions <code>nmc_LGM50_ocp_Chen2020 / graphite_LGM50_ocp_Chen2020</code>)
Nominal capacity	5.085 Ah (calibrated to measured 1C capacity, R5)	<code>`runs/exp/t1_r3/cell/r5_v11_final.json`</code>
Measured 1C capacity	5.0849 Ah (1C CC discharge to 2.5 V, 25 °C)	<code>`runs/exp/t1_r3/cell/r5_v11_1c_spme.json:capacity_ah`</code>
Voltage window	2.5 – 4.2 V	Chen2020: Lower/Upper voltage cut-off
Electrode dimensions	height 65 mm × width 1580 mm (single unwound layer, area 0.1027 m ²)	Chen2020: Electrode height/width
Cell shell dimensions (H×W×T)	Not provided (no shell parameter in the set)	—
Layer stack thickness	184.8 μm (75.6 + 85.2 + 8 + 10 + 6 μm)	<code>`r5_v11_energy.json:thickness_m`</code>
Electrolyte	1 M LiPF ₆ base (Nyman2008 transport model); design transport targets $\sigma=1.8\text{ S/m}$, $D=6.0\text{e-}10\text{ m}^2/\text{s}$, $t_{\pm}=0.55$ (advanced-electrolyte estimate, flagged in plan)	Chen2020 functions + <code>`r5_v11_final.json`</code>
Additive candidates	None evaluated (no Stage-2 molecular funnel in this case; task is cell-design level)	— (honest omission)
Cation transference number	0.55	<code>`r5_v11_final.json`</code>

2. Electrode and Separator

Layer	Thickness	Porosity	Active volume fraction	Particle radius	Density	Source
Positive (NMC811)	75.6 μm	0.45	0.55	0.8 μm	3262 kg/m ³	Chen2020 (thickness/density) + <code>`r5_v11_final.json`</code> (porosity/radius)
Negative (graphite)	85.2 μm	0.42	0.58	0.8 μm	1657 kg/m ³	same
Separator	8 μm	0.60	—	—	397 kg/m ³	same
Positive current collector	10 μm Al	—	—	—	2700 kg/m ³	same
Negative current collector	6 μm Cu	—	—	—	8960 kg/m ³	same

N/P ratio: 0.624 geometric = $(\epsilon_{\text{am-c_max}} \cdot \delta)_{\text{neg}} \div (\epsilon_{\text{am-c_max}} \cdot \delta)_{\text{pos}} = (0.58 \times 33133 \times 85.2\text{ }\mu\text{m}) \div (0.55 \times 63104 \times 75.6\text{ }\mu\text{m})$, per the template formula, from parameter-set keys (c_max values read from pybamm Chen2020 set; $\epsilon_{\text{am}} = 1 - \text{porosity}$).

Caveat: the Chen2020 negative OCP fit saturates above stoich 0.90 (OCP_{neg}(0.90) = OCP_{neg}(1.00) = 0.092 V, mechanically evaluated), so the practical negative window differs from the geometric window; the as-operated N/P is therefore verified by simulation instead: plating-free at true 4C with anode potential min +0.0425 V (``r5_v11_4c.json:anode_potential_v``).

3. Process Design Parameters

Parameter	Value	Formula	Source
Positive areal density	135.63 g/m ²	$\delta \times (1-\epsilon) \times \rho = 75.6\text{e-}6 \times 0.55 \times 3262$	<code>`r5_v11_energy.json:layer_kg_m2_positive_electrode`</code>
Negative areal density	81.88 g/m ²	$\delta \times (1-\epsilon) \times \rho = 85.2\text{e-}6 \times 0.58 \times 1657$	<code>`r5_v11_energy.json:layer_kg_m2_negative_electrode`</code>
Positive compaction density	1.79 g/cm ³	$\rho \times (1-\epsilon) = 3262 \times 0.55 / 1000$	derived from Chen2020 + porosity
Negative compaction density	0.96 g/cm ³	$\rho \times (1-\epsilon) = 1657 \times 0.58 / 1000$	derived from Chen2020 + porosity

Parameter	Value	Formula	Source
Electrolyte fill amount	9.19 g	pore volume $\times$ 1.2 g/cm ³ $\times$ 1.0 = 7.662 cm ³ $\times$ 1.2	pore volume = $A \times (\delta_{\text{pos}} \cdot \epsilon_{\text{pos}} + \delta_{\text{neg}} \cdot \epsilon_{\text{neg}} + \delta_{\text{sep}} \cdot \epsilon_{\text{sep}})$; electrolyte density 1.2 g/cm ³ literature value, fill factor 1.0 idealized
Formation recommendation	0.1C CC to 4.2 V, 25 °C, 2 cycles	design recommended value; actual production-line value requires tuning	—

4. Mass Breakdown

Component	Mass (g)	Source
Positive electrode (active solids)	13.930	`r5_v11_energy.json:layer_kg_m2.positive_electrode` $\times$ 0.1027 m ² $\times$ 1000
Negative electrode (active solids)	8.409	`r5_v11_energy.json:layer_kg_m2.negative_electrode` $\times$ 0.1027 m ² $\times$ 1000
Positive current collector (Al)	2.773	`r5_v11_energy.json:layer_kg_m2.positive_cc` $\times$ 0.1027 m ² $\times$ 1000
Negative current collector (Cu)	5.521	`r5_v11_energy.json:layer_kg_m2.negative_cc` $\times$ 0.1027 m ² $\times$ 1000
Separator	0.130	`r5_v11_energy.json:layer_kg_m2.separator` $\times$ 0.1027 m ² $\times$ 1000
Total (contract caliber, electrolyte excluded)	30.763	`r5_v11_energy.json:mass_kg` (calc-energy contract: electrolyte density absent from parameter set $\rightarrow$ excluded)
Electrolyte (additive)	9.19	see §3 fill amount; NOT in contract mass
Total incl. electrolyte	40.77	sum

5. Performance Verification

Item	Result	Criterion (entry 0)	Verdict	Source
Energy density	605.33 Wh/kg	≥ 392.61 Wh/kg	✓ PASS	`r5_v11_energy.json:energy_density_wh_kg`
Volumetric energy density	981.20 Wh/L	(informational)	—	`r5_v11_energy.json:energy_density_wh_l`
1C discharge capacity	5.0849 Ah	(no capacity threshold registered)	informational	`r5_v11_1c_spme.json:capacity_ah`
4C fast charge acceptance	4.677 Ah = 92.0% of 1C capacity in 13.8 min at 20.34 A (true 4C)	task: "support 4C fast charge"	✓ (no plating + full charge reachable)	`r5_v11_4c.json:capacity_ah` $\times$ nominal 5.085
4C plating determination	anode potential min +0.0425 V (≥ 0 V)	plated = false	✓ PASS	`r5_v11_4c.json:anode_potential_v`
4C maximum temperature	321.2 K (48.0 °C)	$T_{\text{max_K}} \leq 333.15$ K	✓ PASS	`r5_v11_4c.json:T_max_K`
Overcharge to 4.7 V	v reached 4.700 V; T_{max} 298.57 K; thermal-runaway triggered = false	triggered = false; $T_{\text{max}} \leq 333.15$ K	✓ PASS	`r5_v11_oc.json` + `r5_v11_tr.json`
Rated energy	18.622 Wh	—	—	`r5_v11_energy.json:energy_wh`

6. Design Notes

Changes vs the Chen2020 baseline and why (reasoning cited from the evaluate log rounds R1–R6):

- Transport/kinetics pack** (R2 V1 $\rightarrow$ kept in all later variants): electrolyte conductivity 1.8 S/m (constant, replaces the Nyman2008 temperature function — advanced-electrolyte ceiling assumption, flagged in the Stage-1 plan), D 6.0e-10 m²/s, t_0 0.55, porosity 0.45/0.42. Effect: anode potential -0.19 V $\rightarrow$ $+0.005$ V (plating removed), T_{max} 354 K $\rightarrow$ 328 K (R2 evaluate).
- Particle size 0.8 μm** (R4 V9): $2\times$ active surface vs 1.5 μm — widened the 4C anode margin to $+0.0424$ V (R4 evaluate).
- Current collectors 16/12 $\rightarrow$ 10/6 μm** (R4 V9): holds ED far above the 392.61 target despite added porosity mass.
- Thermal: $h = 120$ W/m²/K** (R4 V9): 4C T_{max} 321.0 K; overcharge T_{max} 298.57 K.
- Nominal capacity 5.085 Ah** (R5 V11): calibrated to the measured 1C capacity so the harness 4C protocol ($4\times$ nominal = 20.34 A) is a true 4C — units diagnosis of the earlier apparent "acceptance plateau" (charge capacity_ah is C-rate-normalized duration; true Ah = reported $\times$ nominal).
- N/P lever**: tested in R3 (V7) and found net-negative at 4C (thicker negative lengthens the diffusion path) — not used.